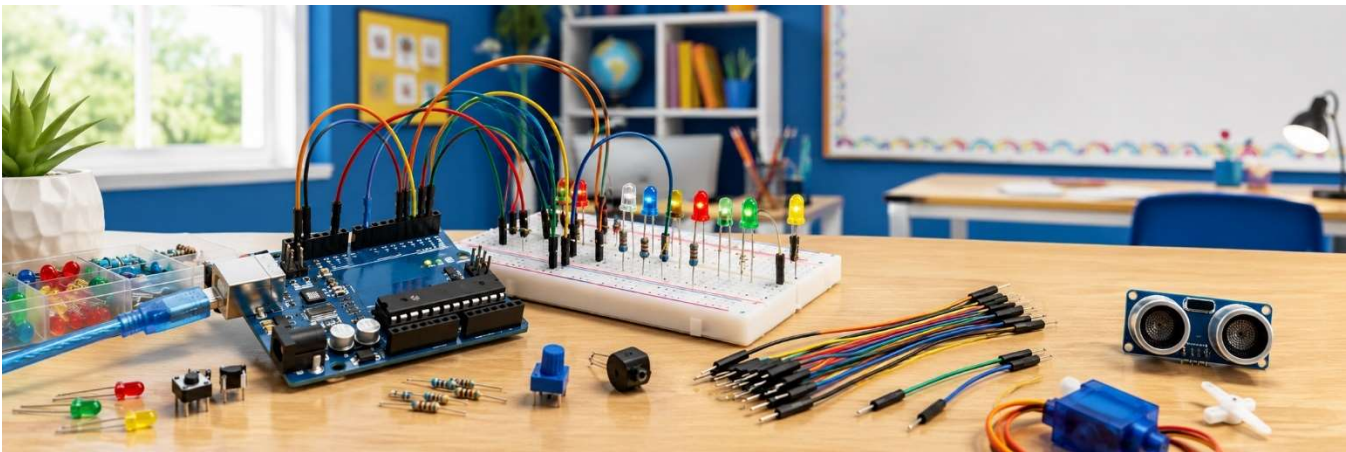


Arduino Beginner STEM Curriculum

Level 1 | Readiness + Project Pack

Grades 5-9 | 7 Projects | Readiness Bootcamp | Capstone Challenge



7 Main Projects

6 Readiness Lessons

Complete Assessments

The Essential Prerequisite Before:

Arduino Intermediate STEM Curriculum | Level 2 | Grades 5-9

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Premium STEM Curriculum Edition

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Use this guide to quickly locate teacher materials, readiness lessons, student pages, projects, assessments, rubrics, and answer keys.

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LEVEL 1 READINESS BOOTCAMP

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TEACHER QUICK START GUIDE

Welcome!

This resource is designed to be picked up and used immediately. Follow the 5 steps below to get started with your Arduino beginner class.

Step	Action
Step 1 — Read This Guide (15 min)	Skim the Teacher Guide and Implementation Notes. Review the Unit Structure at a Glance. Identify which pacing option fits your schedule: 2-week, 4-week, or 6-week.
Step 2 — Gather Materials	Print the Complete Materials List and verify you have all components. Order any missing items at least 2 weeks in advance.
Step 3 — Prep the Classroom	Install Arduino IDE on all student computers. Test-build each circuit before class. Pre-sort components into labeled bins by project.
Step 4 — Print Student Materials	Print one copy of each Student Reading Page, Worksheet, and Exit Ticket per student. Print rubrics for your reference.
Step 5 — Begin Readiness Bootcamp	Do NOT skip the Level 1 Readiness Bootcamp. Even students who have used computers before need these 6 lessons to succeed with Arduino.

Teacher Tip

The biggest mistake is moving too fast. Students need time to rewire, re-upload, and debug. Build in at least 10 extra minutes per project for troubleshooting.

WHAT THIS LEVEL 1 CURRICULUM PREPARES STUDENTS FOR

This Level 1 Beginner Pack is the official prerequisite for the Arduino Intermediate STEM Curriculum (Level 2). It provides a structured, project-based pathway from absolute beginner to a student who can independently build and program Arduino circuits.

LEVEL 1 → LEVEL 2 SKILL BRIDGE

Level 1 Builds This Skill	Level 2 Uses This Skill
Blinking an LED with digitalWrite()	Controlling multiple LEDs in sequences
Reading buttons with digitalRead()	Multi-button input systems
Reading analog sensors with analogRead()	Sensor arrays and data logging
Serial Monitor debugging	Serial plotter and advanced data output
Basic if/else threshold logic	Nested conditionals and compound logic
tone() with buzzer	Multi-note songs and alarm systems
analogWrite() PWM brightness	PWM motor control and fading effects
Ultrasonic sensor distance	Obstacle-avoiding robot navigation
Servo motor control	Multi-servo robotic arms
millis() introduction	Non-blocking timing and multitasking

Design Philosophy

Each project in this Level 1 pack deliberately introduces a component or concept that reappears in the Level 2 Intermediate Pack. Students are not just completing fun projects — they are building a mental toolkit they will use immediately in Level 2.

COMPLETE MATERIALS LIST

Materials needed per student team (2 students per kit). Quantities listed are per kit unless noted otherwise.

Core Electronics

Component	Quantity per Kit
Arduino Uno R3 (or compatible)	1
USB-A to USB-B cable	1
breadboard	1
Jumper wires (mixed colors, Male-Male)	30+
Red LED	3
Green LED	3
Yellow LED	2
220Ω resistors	10
10kΩ resistors	5
Piezo buzzer (passive)	1
HC-SR04 Ultrasonic sensor	1
Servo motor (SG90 or MG90S)	1
TMP36 temperature sensor	1
Photoresistor (LDR)	2
Push button (momentary, 4-pin)	3
10kΩ potentiometer	1
Multimeter (optional)	1 per class

Classroom Supplies

Item	Notes
Computers with USB ports	One per team minimum
Arduino IDE installed	Free download at arduino.cc
Printed student worksheets	One set per student
Component storage bins	Labeled by component type
Small containers / cups	For holding loose components
Anti-static mat (optional)	Recommended for component safety

Shopping Tip

Arduino starter kits available on Amazon (search "Arduino Uno Starter Kit") often include all required components for \$25–\$35 per kit. Buying individual components in bulk is more economical for larger classes.

NGSS STANDARDS ALIGNMENT

Science & Engineering Practices

- SEP 1: Asking Questions and Defining Problems
- SEP 3: Planning and Carrying Out Investigations
- SEP 4: Analyzing and Interpreting Data
- SEP 5: Using Mathematics and Computational Thinking
- SEP 6: Constructing Explanations and Designing Solutions
- SEP 8: Obtaining, Evaluating, and Communicating Information

Crosscutting Concepts

- CCC 2: Cause and Effect
- CCC 4: Systems and System Models
- CCC 6: Structure and Function

Disciplinary Core Ideas

- MS-ETS1-1: Define the criteria and constraints of a design problem
- MS-ETS1-2: Evaluate competing design solutions
- MS-ETS1-3: Analyze data from tests to determine similarities and differences
- MS-ETS1-4: Develop a model to generate data for iterative testing

Computer Science Connections

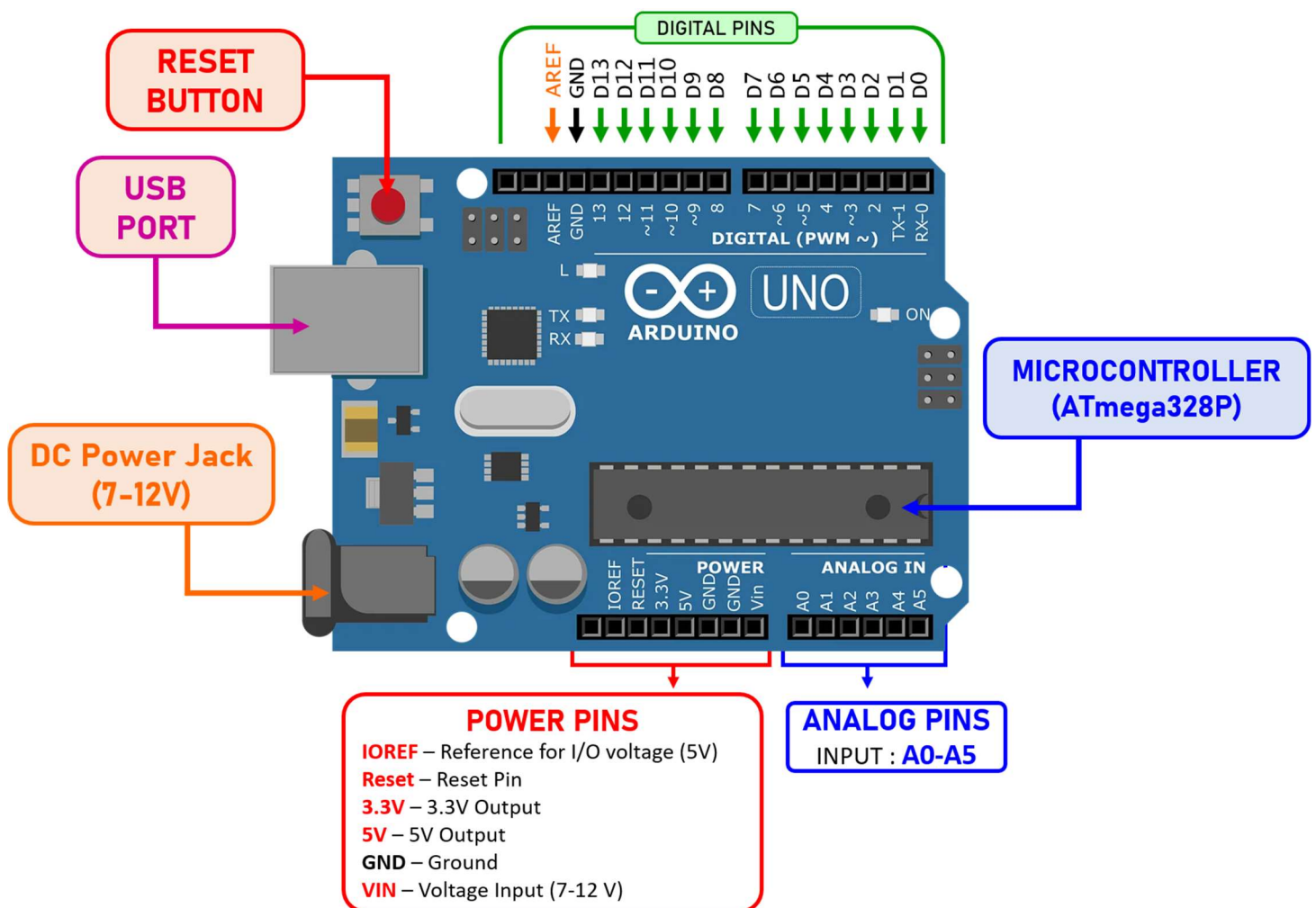
- CSTA K-12 CS Framework: Algorithms & Programming (Level 2 / Grades 6–8)
- Sequence, selection, and repetition in programs
- Variables, data types, and input/output
- Debugging and iterative development

Step-by-Step Activity

- Hold the Arduino Uno and find the USB port. This is how the board connects to a computer for power and programming.
- Locate the digital pins (labeled 0–13). These control LEDs, buttons, buzzers, and more.
- Find the analog pins (labeled A0–A5). These read sensors like temperature or light.
- Find the 5V, 3.3V, and GND pins. These supply power to your components.
- Trace a simple path: 5V → component → GND. Discuss why the path must be complete.
- Label a diagram of the Arduino Uno with all parts identified above.

Code Example

```
// No code this lesson – explore the board!  
// Coming in Lesson 2: uploading your first sketch.
```

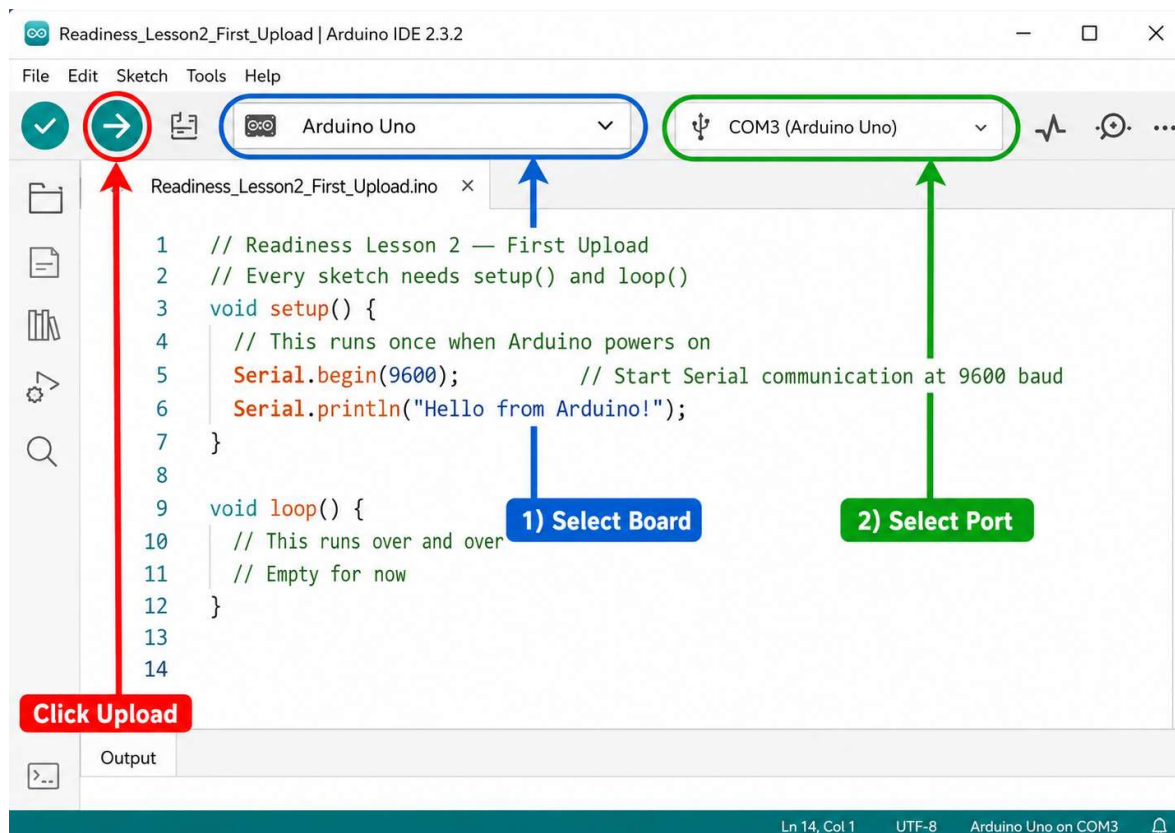


Step-by-Step Activity

- Open Arduino IDE on your computer.
- Go to Tools → Board and select "Arduino Uno".
- Go to Tools → Port and select the correct port (it will show something like COM3 or /dev/ttyUSB0).
- Type or paste the code example below into the sketch window.
- Click the Upload button (right arrow icon).
- Watch the status bar — it should say "Done uploading".
- Open the Serial Monitor (Tools → Serial Monitor or Ctrl+Shift+M).
- Set the baud rate to 9600 at the bottom of the Serial Monitor window.
- You should see "Hello from Arduino!" printed in the Serial Monitor.

Code Example

```
// Readiness Lesson 2 - First Upload
// Every sketch needs setup() and loop()
void setup() {
  // This runs once when Arduino powers on
  Serial.begin(9600);          // Start Serial communication at 9600 baud
  Serial.println("Hello from Arduino!");
}
void loop() {
  // This runs over and over
  // Empty for now
}
```

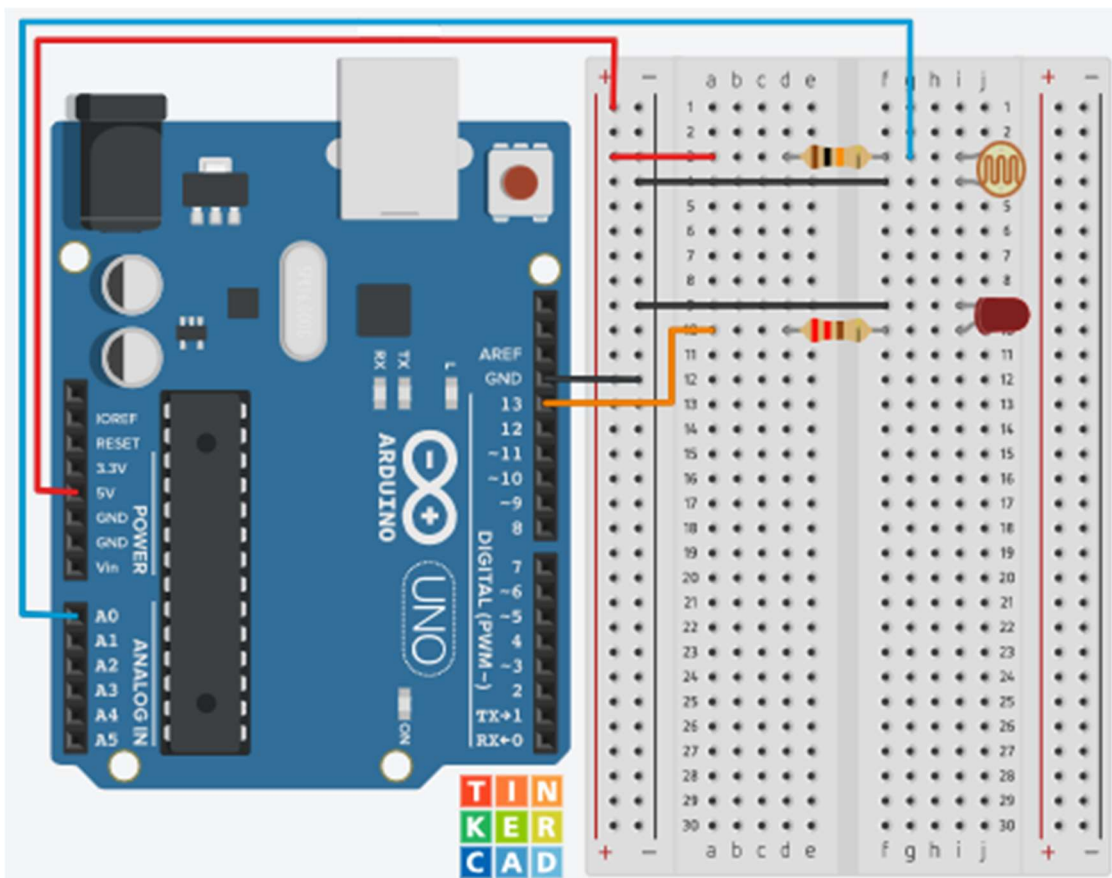


Materials Needed

- Arduino Uno
- Breadboard
- Photoresistor (LDR)
- 10k Ω resistor
- LED
- 220 Ω resistor
- Jumper wires
- USB cable

Step-by-Step Activity

- Wire a photoresistor and 10k Ω resistor in series between 5V and GND (voltage divider — see diagram).
- Connect the middle point (between the resistor and photoresistor) to analog pin A0.
- Wire an LED with a 220 Ω resistor to pin 13.
- Upload the code below.
- Open the Serial Monitor at 9600 baud.
- Cover the photoresistor with your hand and observe the numbers change.
- Adjust the threshold value in the code so the LED turns on only in dim light.
- Record your threshold value in your STEM notebook.



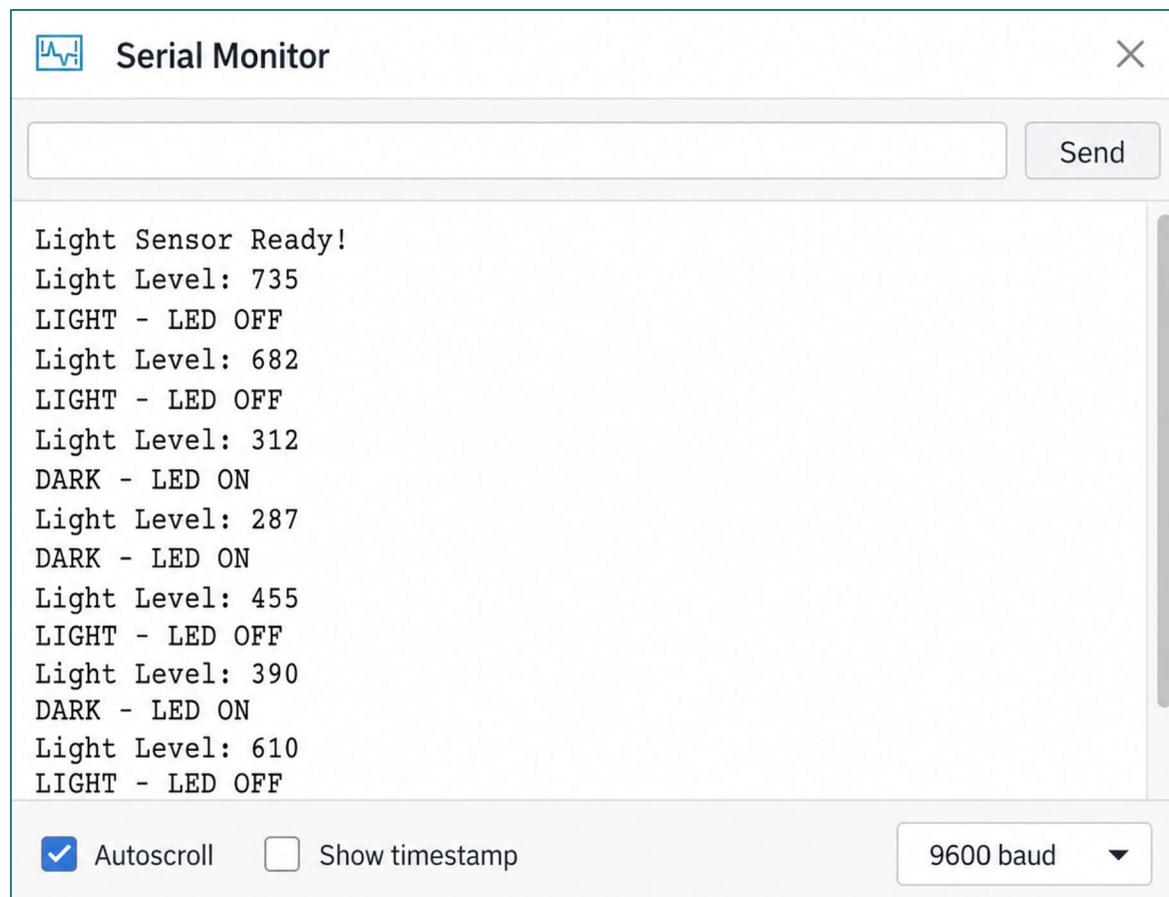
Code Example

```
// Readiness Lesson 6 - Serial Monitor + If/Else
const int LDR_PIN = A0;    // Photoresistor on analog pin A0
const int LED_PIN = 13;
int threshold = 400;       // Adjust this value for your lighting conditions

void setup() {
  pinMode(LED_PIN, OUTPUT);
  Serial.begin(9600);
  Serial.println("Light Sensor Ready!");
}

void loop() {
  int lightValue = analogRead(LDR_PIN); // Read light level (0-1023)
  Serial.print("Light Level: ");
  Serial.println(lightValue);

  if (lightValue < threshold) {          // If it is dark...
    digitalWrite(LED_PIN, HIGH);        // Turn LED on
    Serial.println("DARK - LED ON");
  } else {                              // Otherwise...
    digitalWrite(LED_PIN, LOW);         // Turn LED off
    Serial.println("LIGHT - LED OFF");
  }
  delay(200);
}
```



STUDENT WORKSHEET

Answer the following questions in complete sentences in your STEM notebook:

What is the difference between `Serial.print()` and `Serial.println()`?

Answer: _____

What range of values does `analogRead()` return?

Answer: _____

In the code, what happens if `lightValue` is greater than the threshold?

Answer: _____

What is a threshold and why is it useful in sensor projects?

Answer: _____

EXIT TICKET

What is the difference between `Serial.print()` and `Serial.println()`?

PROJECT 1: ULTRASONIC DISTANCE SENSOR

Measure distance with an HC-SR04 sensor — just like a parking sensor!

Learning Objectives

- Use the HC-SR04 ultrasonic sensor to measure distance in centimeters
- Apply the pulseIn() function to read echo time
- Convert pulse time to distance using a formula
- Use threshold logic to trigger an LED alert
- Display real-time distance readings on the Serial Monitor

Key Vocabulary

Term	Definition
HC-SR04	An ultrasonic distance sensor that sends a sound pulse and measures how long it takes to bounce back.
Trig Pin	The trigger pin — sends a short ultrasonic pulse.
Echo Pin	Receives the returning pulse and allows duration measurement.
pulseIn()	Measures how long a pulse is in microseconds.
Distance Formula	$\text{distance (cm)} = \text{duration} \times 0.034 / 2$
Threshold	A set distance value — getting closer than this triggers an alert.

Materials Needed

- Arduino Uno • HC-SR04 ultrasonic sensor • Red LED • 220Ω resistor • Breadboard
- Jumper wires • USB cable

TEACHER NOTE

Project 1 introduces the Ultrasonic Distance Sensor. Walk through the circuit diagram before students build. Verify all circuits before uploading code. Common issues are listed in the troubleshooting table below.

Capstone Project Rubric

Criteria	4 — Exceeds	3 — Meets	2 — Approaching	1 — Needs Support
Problem Statement	Clear, specific problem statement with real-world context	Problem statement is clear and complete	Problem statement is present but vague	Problem statement missing or unclear
Circuit Building	Circuit is correct, organized, and includes extensions	Circuit is correctly built and functional	Circuit works with teacher assistance	Circuit not functional
Code Functionality	Code works fully and includes student-added improvements	Code works as designed	Code partially works; some features incomplete	Code does not upload or run
Testing & Iteration	Multiple test iterations documented with clear improvements	Testing table completed with evidence of iteration	Some testing documented; limited iteration	No testing documentation
Presentation	Clear, confident explanation with real-world connection	Explanation is clear; covers circuit, code, and function	Explanation covers most required points	Explanation incomplete or unclear

Mid-Unit Quiz

TEACHER NOTE

Administer after Project 3 or 4. Covers Readiness Lessons and first half of projects.

Name: _____ Date: _____ Period: _____

Circle the best answer:

1. Which function reads a digital pin?

- A. analogRead()
- B. digitalRead()
- C. Serial.print()
- D. pinMode()

2. What value does INPUT_PULLUP read when button is NOT pressed?

- A. LOW / 0
- B. HIGH / 1
- C. 512
- D. 1023

3. What does map(512, 0, 1023, 0, 180) return?

- A. 0
- B. 90
- C. 180
- D. 512

4. Which pin mode is needed to use analogWrite()?

- A. INPUT
- B. OUTPUT
- C. INPUT_PULLUP
- D. PWM_MODE

5. What does millis() return?

- A. Minutes since start
- B. Seconds since start
- C. Milliseconds since start
- D. Microseconds since start